

Electromagnetics I

Lecture 5: Application of Maxwell's Equations

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Office Hours: Thursday, 8:00 AM - 9:00 AM, MSE 016

Outline

- Q&A
- Gauss's law for electric field
- Gauss's law for magnetic field
- Faraday's law
- Ampere's law
- Classical electromagnetism and special relativity

The angle between two vectors



Find the angle between $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 7 \end{pmatrix}$

1. Find the dot product

$$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y$$

$$\mathbf{a} \cdot \mathbf{b} = (3)(1) + (-2)(7)$$

$$\mathbf{a} \cdot \mathbf{b} = -11$$

3. Substitute into the angle formula

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|}$$

$$\cos \theta = \frac{-11}{\sqrt{13} \sqrt{50}}$$

2. Find the magnitude of the vectors

$$|\mathbf{a}| = \sqrt{3^2 + (-2)^2} = \sqrt{13}$$

$$|\mathbf{b}| = \sqrt{1^2 + 7^2} = \sqrt{50}$$

4. Use inverse cosine for the angle

$$\theta = \cos^{-1}\left(\frac{-11}{\sqrt{13} \sqrt{50}}\right)$$

$$\theta = 116^\circ$$

Lorentz Force

23. A charged particle (charge q , mass m) is moving under the influence of a magnetic field

$$\mathbf{B} = B_0 \mathbf{a}_z.$$

If the velocity vector of the particle is

$$\mathbf{v} = v_x \mathbf{a}_x + v_y \mathbf{a}_y + v_z \mathbf{a}_z$$

Show that the acceleration of this particle lies in the x - y plane.

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

$\mathbf{F}$ = force

q = electric charge

$\mathbf{E}$ = external electric field

$\mathbf{v}$ = velocity

$\mathbf{B}$ = magnetic field

Maxwell's Equations

INTEGRAL FORMS

$$\oint_s \epsilon_o \epsilon_r \mathbf{E} \cdot d\mathbf{s} = \int_v \rho_v dv$$

$$\oint_s \mathbf{B} \cdot d\mathbf{s} = 0$$

$$\oint_c \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d}{dt} \int_s \mathbf{B} \cdot d\mathbf{s}$$

$$\oint_c \mathbf{H} \cdot d\boldsymbol{\ell} = \int_s \mathbf{J} \cdot d\mathbf{s} + \frac{d}{dt} \int_s \epsilon_o \epsilon_r \mathbf{E} \cdot d\mathbf{s}$$

DIFFERENTIAL FORMS

$$\boldsymbol{\nabla} \cdot \epsilon_o \epsilon_r \mathbf{E} = \rho_v \quad (4.1)$$

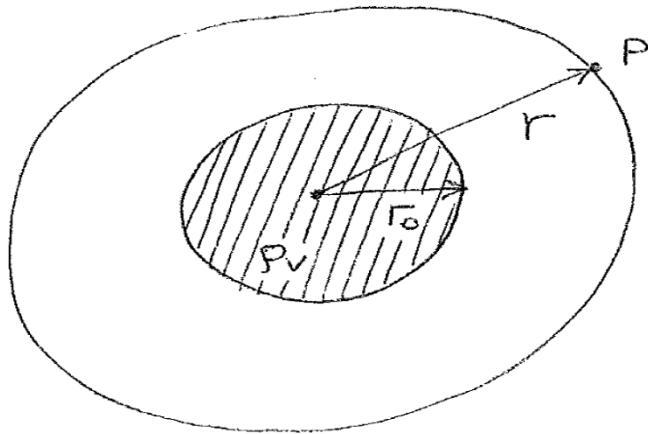
$$\boldsymbol{\nabla} \cdot \mathbf{B} = 0 \quad (4.2)$$

$$\boldsymbol{\nabla} \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (4.3)$$

$$\boldsymbol{\nabla} \times \mathbf{H} = \mathbf{J} + \frac{\partial \epsilon_o \epsilon_r \mathbf{E}}{\partial t} \quad (4.4)$$

Classical Example

ρ_v - charge density



$$\oint_S \epsilon_0 \vec{E} \cdot d\vec{S} = \int_V \rho_v dV$$

case: $r > r_0$

$$\text{Total flux} = \oint_S \epsilon_0 E_r \vec{a}_r \cdot d\vec{S}_r$$

We need element of area $d\vec{S}_r$
[we look page 15]

$$\vec{S}_r = r^2 \sin \theta d\theta d\phi \vec{a}_r$$

$$\begin{aligned} \theta &\in (0; \pi) \\ \phi &\in (0; 2\pi) \end{aligned}$$

$$\text{Total flux} = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \epsilon_0 E_r \vec{a}_r \cdot r^2 \sin \theta d\theta d\phi \vec{a}_r =$$

$$\therefore \text{Total Flux} = \epsilon_0 E_r r^2 \int_{\phi=0}^{2\pi} d\phi \int_{\theta=0}^{\pi} \sin\theta d\theta =$$

$$= 2\pi \epsilon_0 E_r r^2 (-\cos\theta) \Big|_0^{\pi} = 4\pi \epsilon_0 E_r \cdot r^2$$

Charge enclosed by the surface :

$$Q = \int_V \rho_v dV \quad [\text{We need element of volume — look page 15}]$$

$$dV = r^2 \sin\theta dr d\theta d\phi$$

$$Q = \int_V \rho_v \cdot r^2 \sin\theta dr d\theta d\phi =$$

$$= \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \int_{r=0}^{r_0} \rho_v r^2 \sin\theta dr d\theta d\phi =$$

$$= 2\pi \cdot 2 \cdot \rho_v \cdot \frac{r_0^3}{3} = \frac{4}{3} \pi \rho_v r_0^3$$

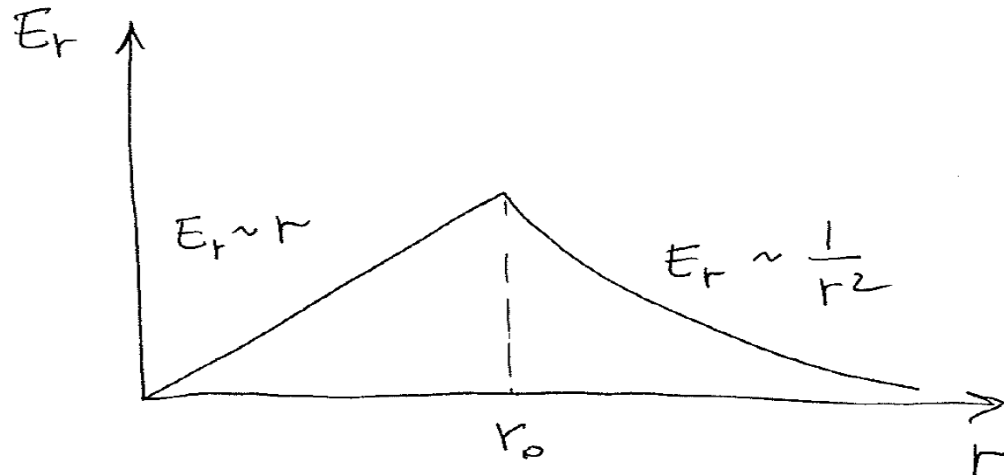
$$4\pi\epsilon_0 E_r \cdot r^2 = \frac{4}{3}\pi\rho_v \cdot r^3$$

$$\Rightarrow \boxed{\begin{aligned} E_r &= \frac{\rho_v r}{3\epsilon_0} \quad (\text{V/m}) \quad r < r_0 \\ \vec{E} &= E_r \vec{a}_r \end{aligned}} \quad (2)$$

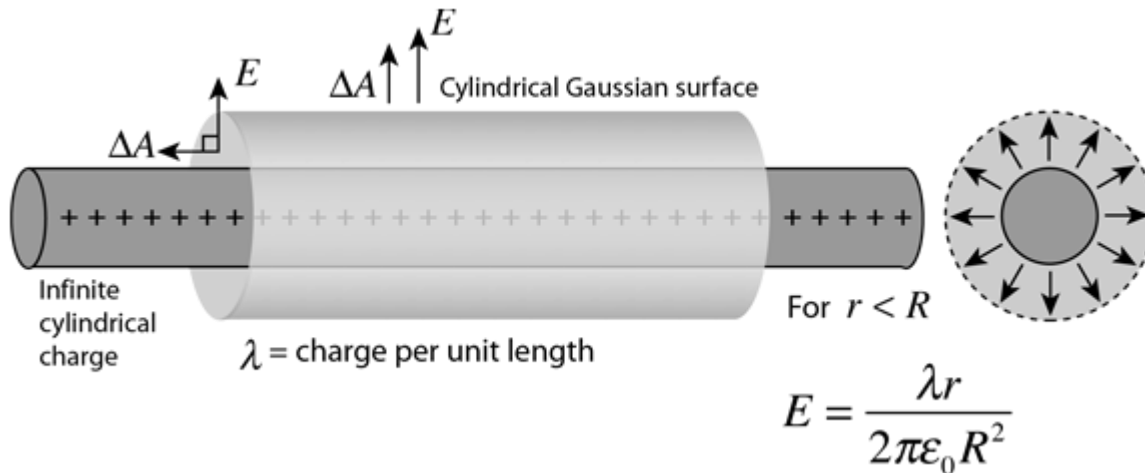
Check boundary:

$$E_{r=r_0} = \frac{\rho_v r_0}{3\epsilon_0} \quad (\text{from (2)})$$

$$\text{From (1): } E_{r=r_0} = \frac{\rho_v r_0^3}{3\epsilon_0 r_0^2} = \frac{\rho_v r_0}{3\epsilon_0} \quad !!!$$



Electric Field: Cylinder of Charge



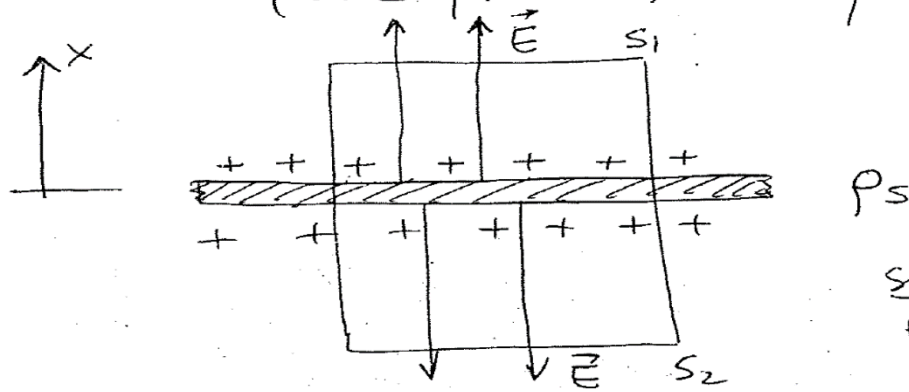
$$\Phi = E 2\pi r L = \frac{\lambda L}{\epsilon_0}$$

For $r \geq R$

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

This expression is a good approximation for the field close to a long cylindrical charge.

Infinitely Large Plane Surface (one plate of a capacitor)



surface: rectangular
parallelepiped

$$\vec{E}_{\text{front}} = E_x \cdot \vec{a}_x$$

$$\oint_S \epsilon_0 \vec{E} \cdot d\vec{S} = Q$$

$$\int_{S_1} \epsilon_0 E_x \vec{a}_x \cdot A \vec{a}_x +$$

$$+ \int_{S_2} \epsilon_0 (-E_x \vec{a}_x) \cdot (-A \vec{a}_x) = Q$$

$$2\epsilon_0 E_x \cdot A = \rho_s \cdot A$$

$$\Rightarrow E_x = \frac{\rho_s}{2\epsilon_0}$$

$$\vec{E} = E_x \vec{a}_x = \frac{\rho_s}{2\epsilon_0} \vec{a}_x$$

for $x > 0$

$$\vec{E} = -\frac{\rho_s}{2\epsilon_0} \vec{a}_x, \quad x < 0$$

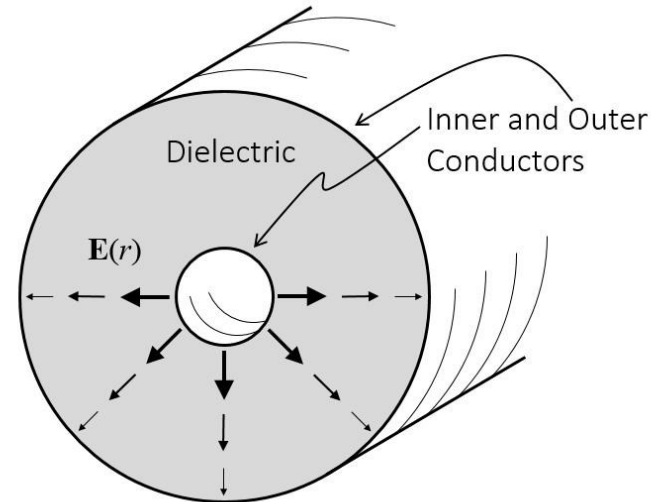
Electric field inside the charged coaxial cable (static)

Problem setup

Consider a coaxial cable with:

- inner conductor radius: a
- outer conductor inner radius: b
- charge per unit length on inner conductor: $+\lambda$
- charge per unit length on outer conductor: $-\lambda$

We want the electric field $E(r)$ in different regions.



Symmetry

1. Symmetry

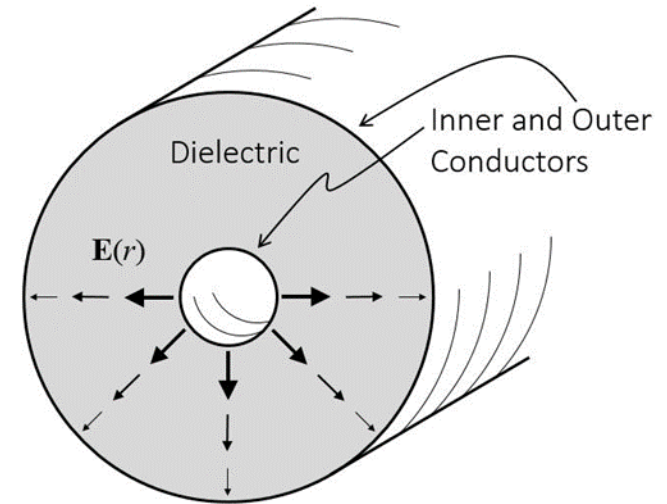
A coaxial cable has **cylindrical symmetry**.

So the electric field must be:

- purely radial
- depends only on distance r from the axis

Thus,

$$\mathbf{E}(r) = E(r) \mathbf{a}_r$$



2. Choose a Gaussian surface

Take a cylindrical Gaussian surface of radius r and length L , coaxial with the cable.

Its curved surface area is:

$$A = 2\pi rL$$

3. Apply Gauss's law

Gauss's law says:

$$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

Since $\mathbf{E}$ is radial and constant on the curved surface:

$$\oint \mathbf{E} \cdot d\mathbf{A} = E(r) (2\pi rL)$$

So:

$$E(r)(2\pi rL) = \frac{Q_{\text{enc}}}{\epsilon_0}$$

Now we evaluate Q_{enc} in each region.

Region 1: Inside the inner conductor ($r < a$)

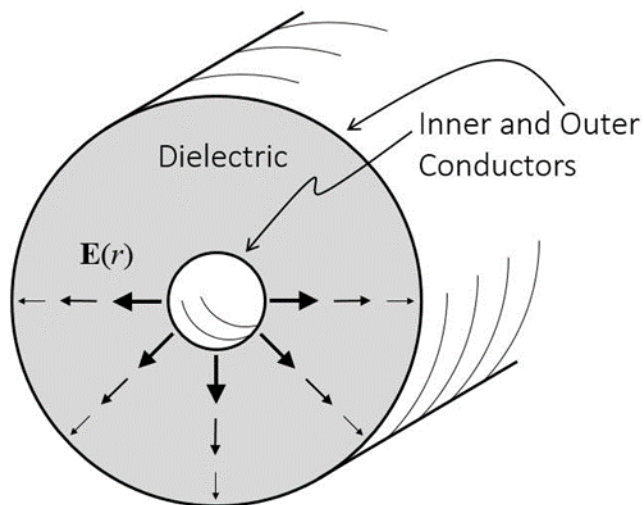
If the inner conductor is a conductor in electrostatic equilibrium, then:

$$E = 0$$

because the electric field inside a conductor is zero.

So:

$$\mathbf{E} = 0, \quad r < a$$



Region 2: Between the conductors ($a < r < b$)

Now the Gaussian cylinder encloses the entire charge on the inner conductor:

$$Q_{\text{enc}} = \lambda L$$

Substitute into Gauss's law:

$$E(r)(2\pi r L) = \frac{\lambda L}{\epsilon_0}$$

Cancel L :

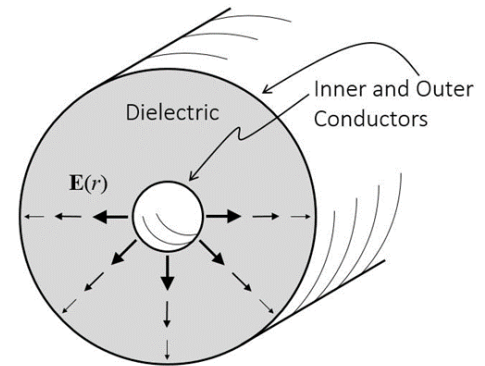
$$E(r)(2\pi r) = \frac{\lambda}{\epsilon_0}$$

Solve for $E(r)$:

$$E(r) = \frac{\lambda}{2\pi\epsilon_0 r}$$

Direction is radial outward if $\lambda > 0$, so:

$$\mathbf{E}(r) = \frac{\lambda}{2\pi\epsilon_0 r} \mathbf{a}_r, \quad a < r < b$$



Region 3: Inside the outer conductor

Inside the metal of the outer conductor, electrostatic equilibrium gives:

$$E = 0$$

So if the outer conductor has thickness, field in the metal is zero.

Region 4: Outside the cable

If the outer conductor carries charge $-\lambda$, then total enclosed charge is:

$$Q_{\text{enc}} = \lambda L + (-\lambda L) = 0$$

So Gauss's law gives:

$$E(r)(2\pi r L) = 0$$

Therefore:

$$\mathbf{E} = 0$$

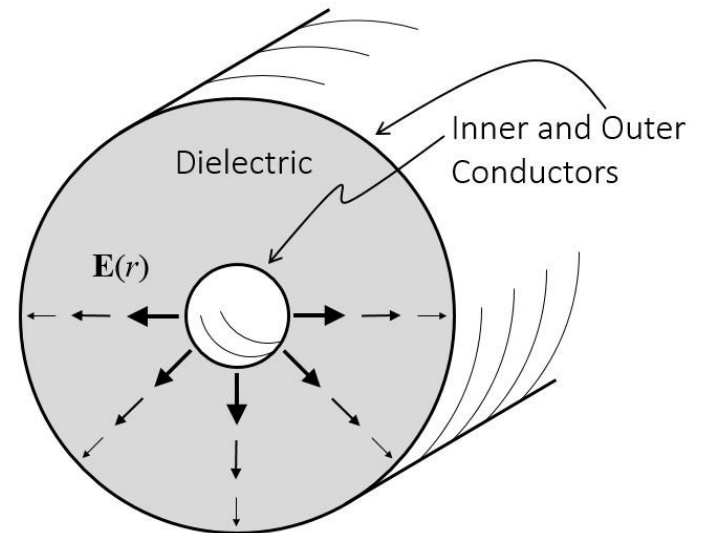
outside the cable.

Final result

$$\mathbf{E}(r) = \begin{cases} 0, & r < a \\ \frac{\lambda}{2\pi\epsilon_0 r} \mathbf{a}_r, & a < r < b \\ 0, & \text{inside conductor and outside the cable} \end{cases}$$

Why this makes sense

- Inside a conductor: no electric field
- Between conductors: field spreads radially
- Outside: total enclosed charge is zero, so no field



Extra: potential difference between conductors

This is often useful for coaxial cables.

$$V = - \int_a^b \mathbf{E} \cdot d\mathbf{l}$$

Since $d\mathbf{l} = dr \mathbf{a}_r$,

$$V = - \int_a^b \frac{\lambda}{2\pi\epsilon_0 r} dr$$

$$V = - \frac{\lambda}{2\pi\epsilon_0} \int_a^b \frac{dr}{r}$$

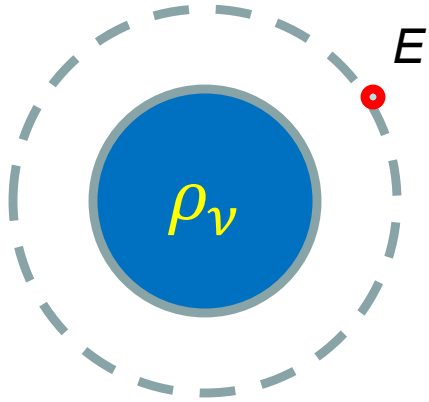
$$V = - \frac{\lambda}{2\pi\epsilon_0} \ln \frac{b}{a}$$

Usually we quote the magnitude:

$$|V| = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{b}{a}$$

Applying Gauss's Law

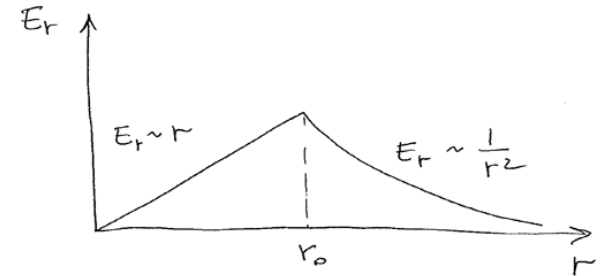
Uniformly
charged sphere



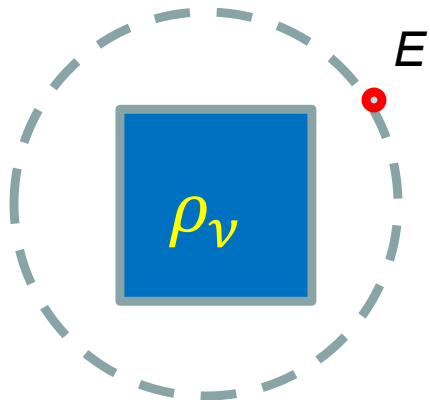
Flux emanating from the surface

$$\epsilon_0 \oint_S \vec{E} \cdot d\vec{S} = \int_V \rho_v dV$$

Electric Field at each point



Uniformly
charged box

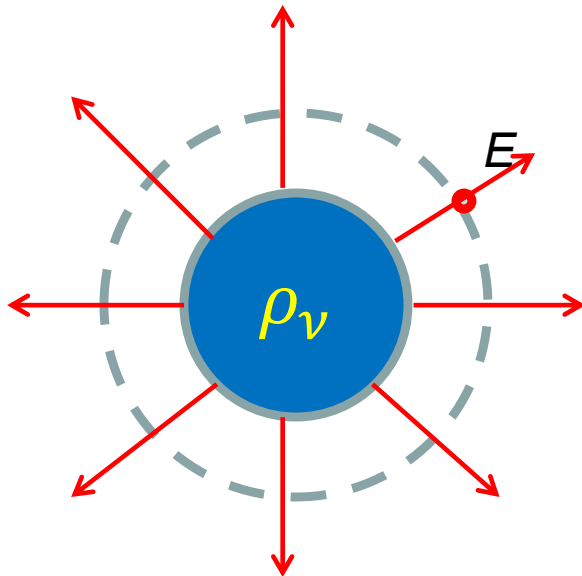


$$\epsilon_0 \oint_S \vec{E} \cdot d\vec{S} = \int_V \rho_v dV$$

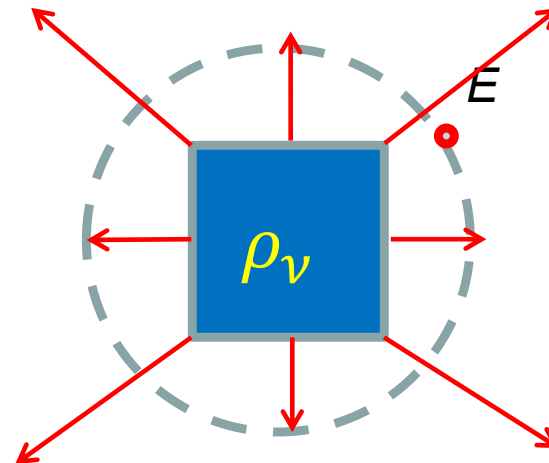
Need to calculate!

Important: Symmetry

Uniformly
charged sphere

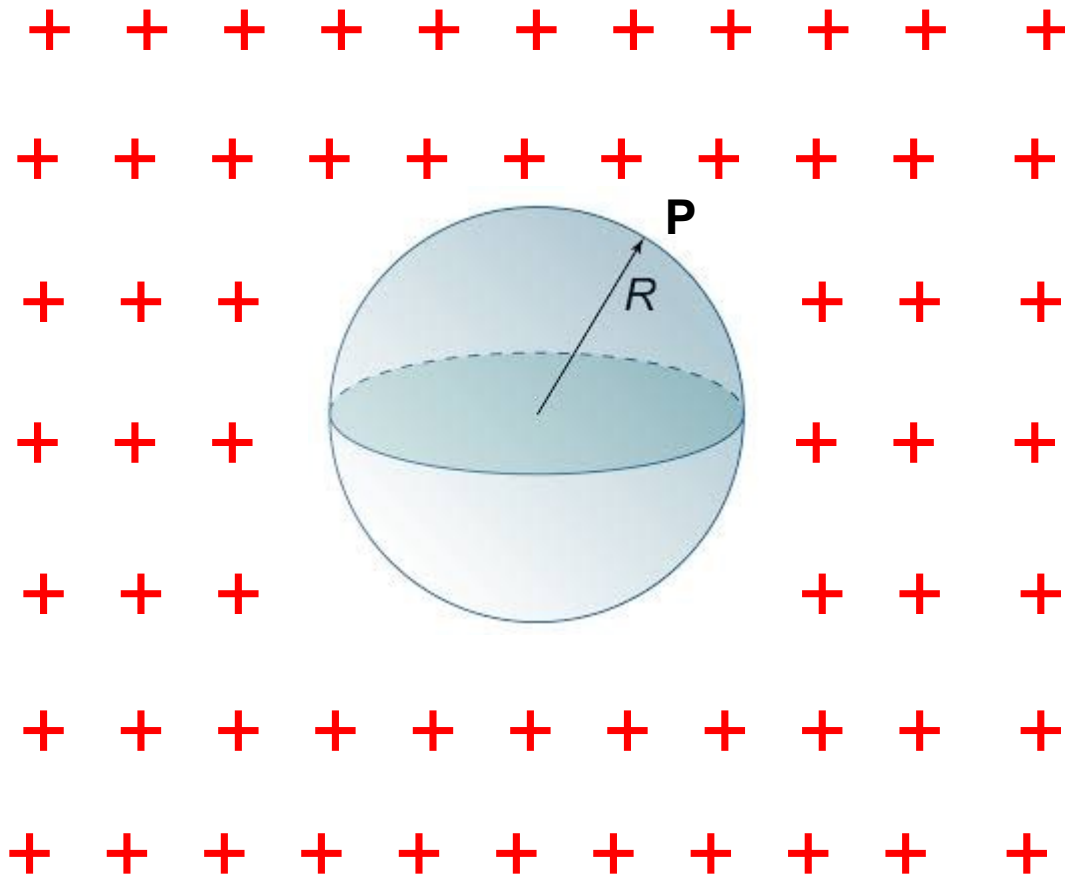


Uniformly
charged box



$$\epsilon_0 \oint_S \vec{E} \cdot d\vec{S} = \int_V \rho_v dV$$

Retrieval Session



$$\epsilon_0 \oint \vec{E} \cdot d\vec{s} = 0$$

$$\text{div} \vec{E} @ P = 0$$

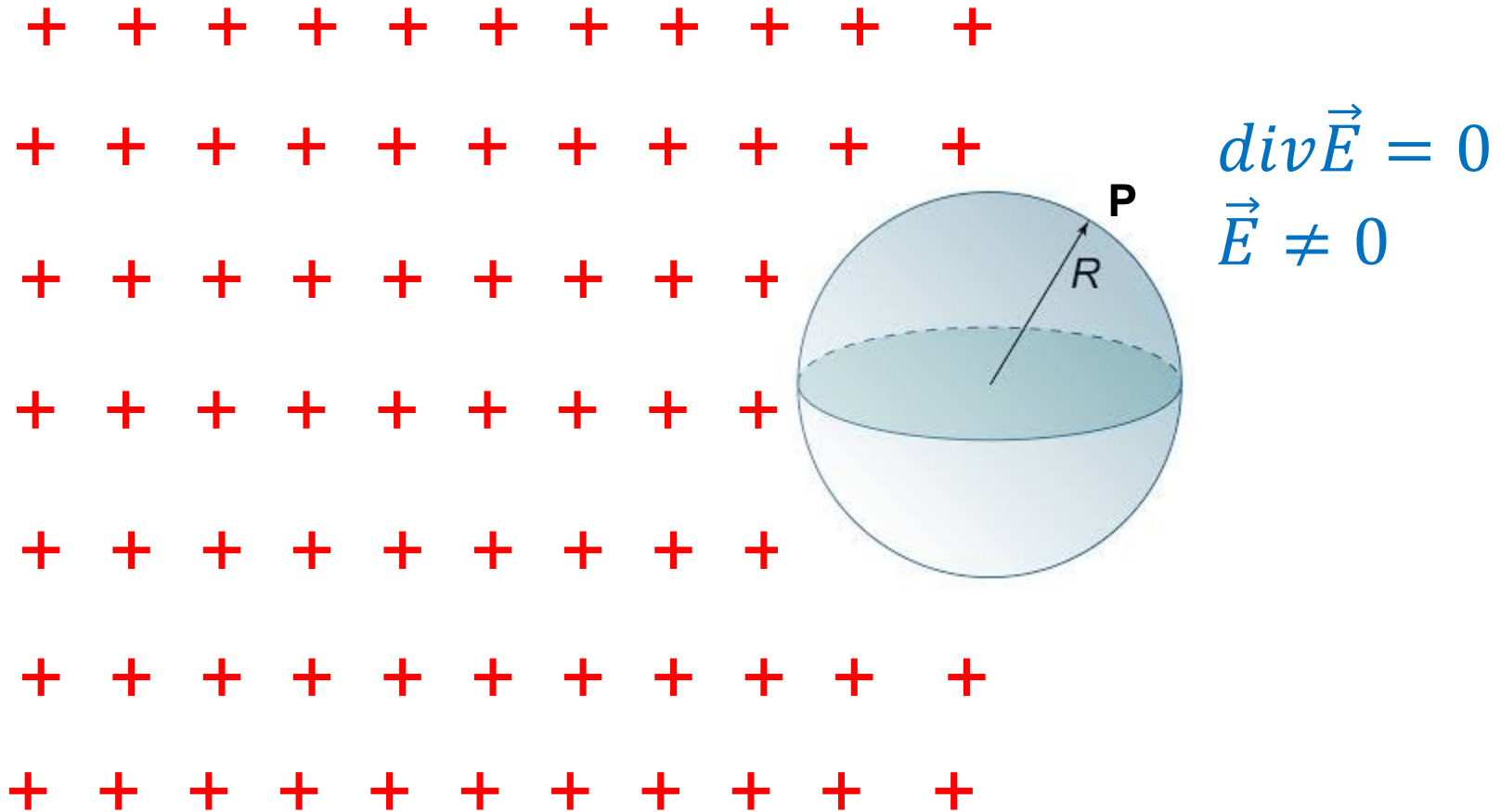
$$\vec{E} = 0$$

Flux through the surface

Divergence at point P: 0 or Not 0

Electric Field at point P: 0, or Not 0, or N/A

Retrieval Session

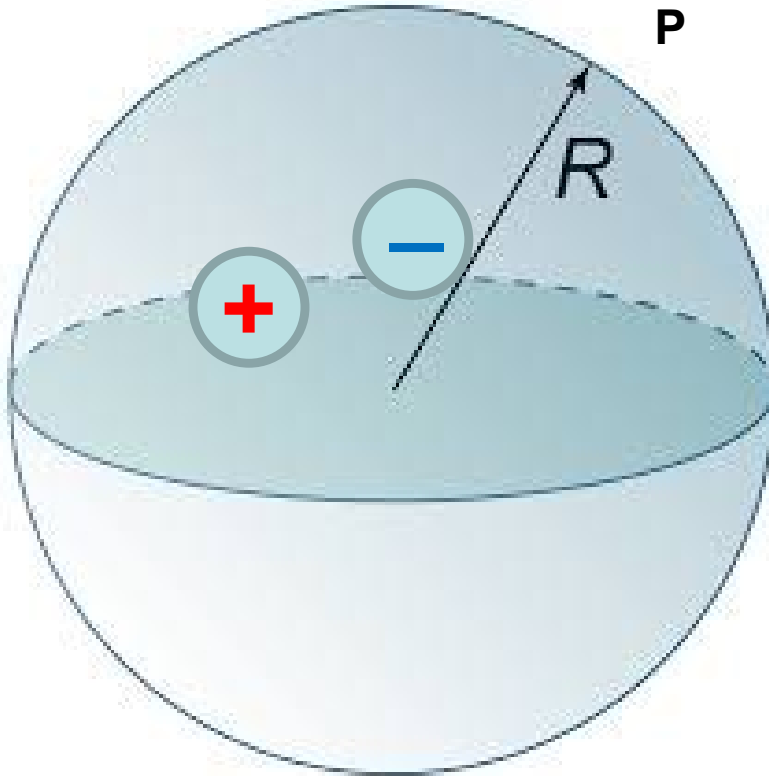


Divergence from the surface: 0 or Not 0

Electric Field at point P: 0, or Not 0, or N/A

Retrieval Session

$$Q = 0$$



$$\epsilon_0 \oint \vec{E} \cdot d\vec{s} = 0$$

$$\text{div} \vec{E} @ P = 0$$
$$\vec{E} (?)$$

Flux through the surface

Divergence at point P: 0 or Not 0

Electric Field at point P: 0, or Not 0, or N/A

Coulomb's Law

$$\mathbf{F} = k \frac{Q_1 Q_2}{R^2} \mathbf{a}_{12}$$

$$k = \frac{1}{4\pi\epsilon_0}$$

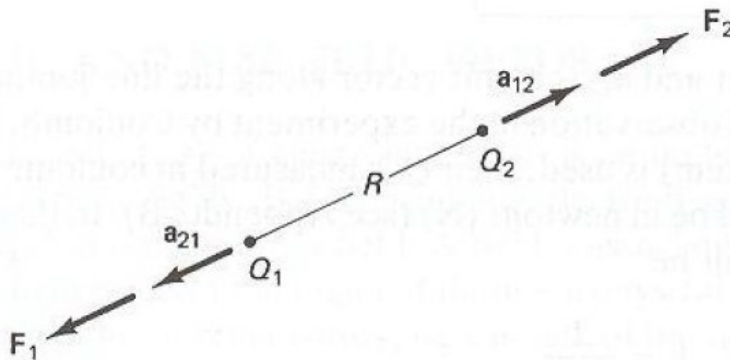


Figure 1.22 The electric force between two point charges Q_1 and Q_2 .

where ϵ_0 is called the permittivity of air (vacuum) and has a value measured in farads per meter (F/m),

$$\epsilon_0 = 8.854 \times 10^{-12} \approx \frac{1}{36\pi} \times 10^{-9} \text{ F/m}$$

Electric Field Superposition

$$\mathbf{E} = \sum_{i=1}^N \frac{Q_i}{4\pi\epsilon_0 R_i^2} \mathbf{a}_{R_i}$$

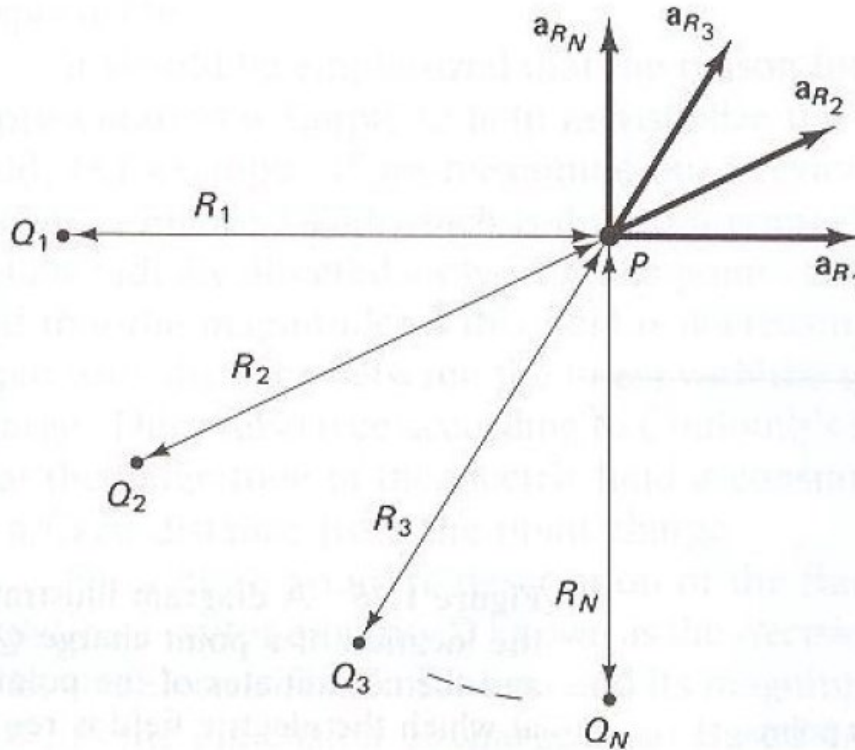
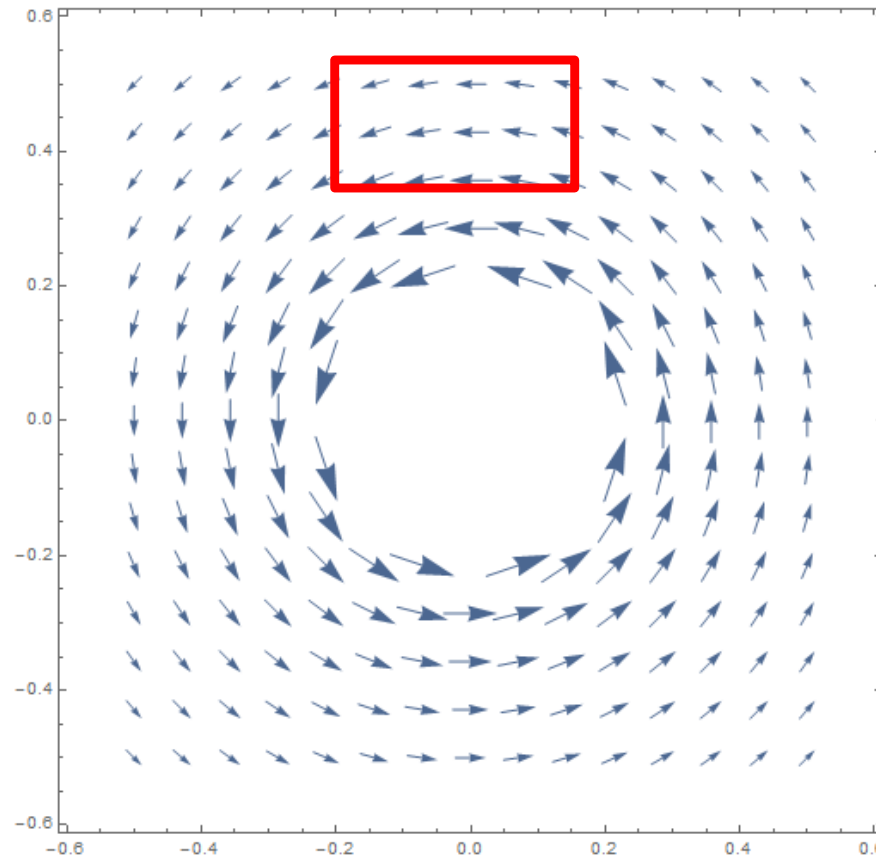


Figure 1.24 The total electric field intensity at point P owing to N point charges equals the vector sum of the electric field intensities owing to all of the charges.

Retrieval Session

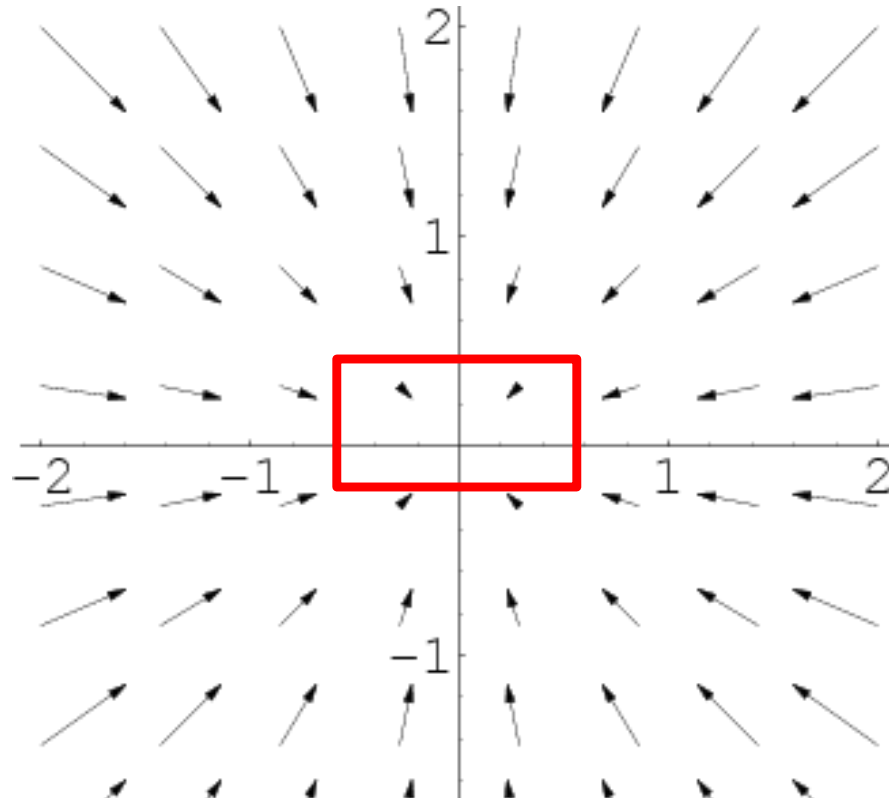
Is it a divergence-free field?



Please type in the chat: **Y** or **N**

Retrieval Session

Is it a divergence-free field?



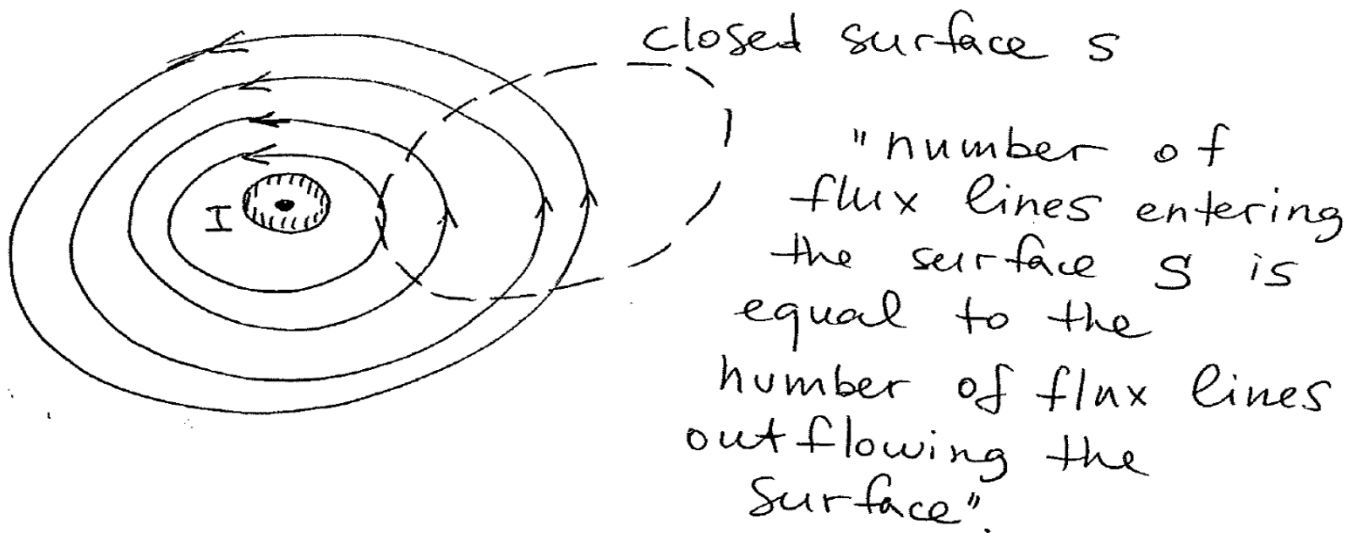
Please type in the chat: **Y** or **N**

What is the sign of the charge in the center? **+** or **-**

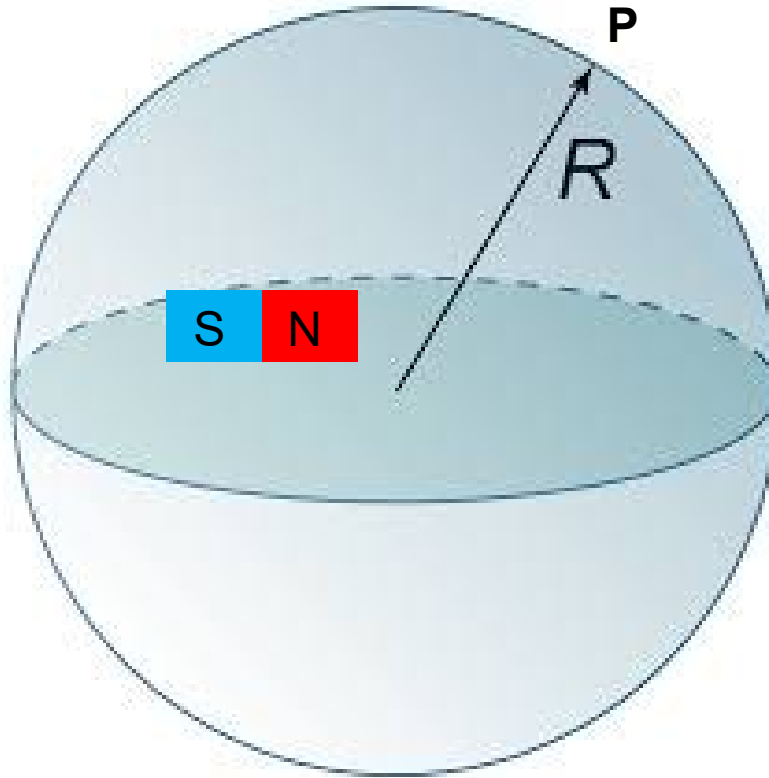
Gauss' Law for Magnetic Field

$$\oint_S \vec{B} \cdot d\vec{S} = 0$$

Total magnetic flux through the closed surface should be equal to zero



Retrieval Session



$$\oint \vec{B} \cdot d\vec{s} = 0$$

$$\text{div} \vec{B} @ P = 0$$

$$\vec{B} \neq 0$$

Flux through the surface

Divergence at point P: 0 or Not 0

Magnetic Field at point P: 0, or Not 0, or N/A

Retrieval Session

- Which of the following vectors may represent a static magnetic field ?

- (A) $\vec{B} = -x\vec{a}_x$
- (B) $\vec{B} = -x\vec{a}_x - y\vec{a}_y$
- (C) $\vec{B} = -x\vec{a}_x + z\vec{a}_z$
- (D) $\vec{B} = -y\vec{a}_x + z\vec{a}_z$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \cdot \mathbf{B} = \frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z}$$

Retrieval Session

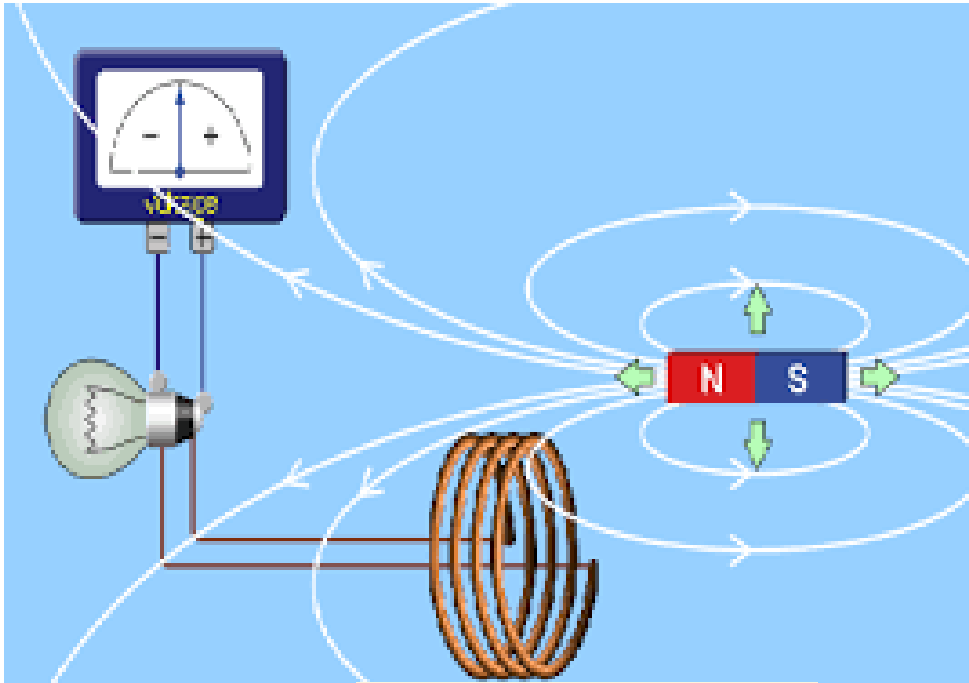
- Which of the following vectors may represent a static magnetic field ?

- (A) $\vec{B} = -x\vec{a}_z$
- (B) $\vec{B} = -2x\vec{a}_x - 2y\vec{a}_y$
- (C) $\vec{B} = -3x\vec{a}_x + 3z\vec{a}_y$
- (D) $\vec{B} = y\vec{a}_y + z\vec{a}_z$

$$\nabla \cdot \mathbf{B} = 0$$

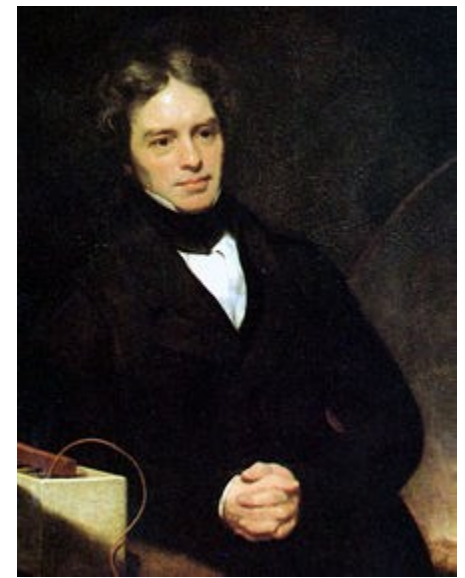
$$\nabla \cdot \mathbf{B} = \frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z}$$

Faraday's Law

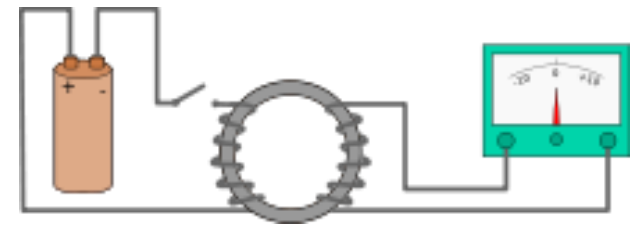


$$\mathcal{E} = - \frac{\Delta \Phi_B}{\Delta t}$$

Faraday's law of induction (shortly called Faraday's law throughout this document) is a basic law of electromagnetism predicting how a magnetic field will interact with an electric circuit to produce an electromotive force (EMF)—a phenomenon called electromagnetic induction. It is the fundamental operating principle of transformers, inductors, and many types of electrical motors, generators and solenoids.



Michael Faraday (22 September 1791
– 25 August 1867)



Different types of EMF

1. A time-varying magnetic field linking a stationary loop; the induced emf is then called the *transformer emf*, $V_{\text{emf}}^{\text{tr}}$.
2. A moving loop with a time-varying surface area (relative to the normal component of $\mathbf{B}$) in a static field $\mathbf{B}$; the induced emf is then called the *motional emf*, $V_{\text{emf}}^{\text{m}}$.
3. A moving loop in a time-varying field $\mathbf{B}$.

The total emf is given by

$$V_{\text{emf}} = V_{\text{emf}}^{\text{tr}} + V_{\text{emf}}^{\text{m}},$$

Stationary Loop in Time-Varying $\mathbf{B}$

It is important to remember that $\mathbf{B}_{\text{ind}}$ serves to oppose the change in $\mathbf{B}(t)$, and not necessarily $\mathbf{B}(t)$ itself.

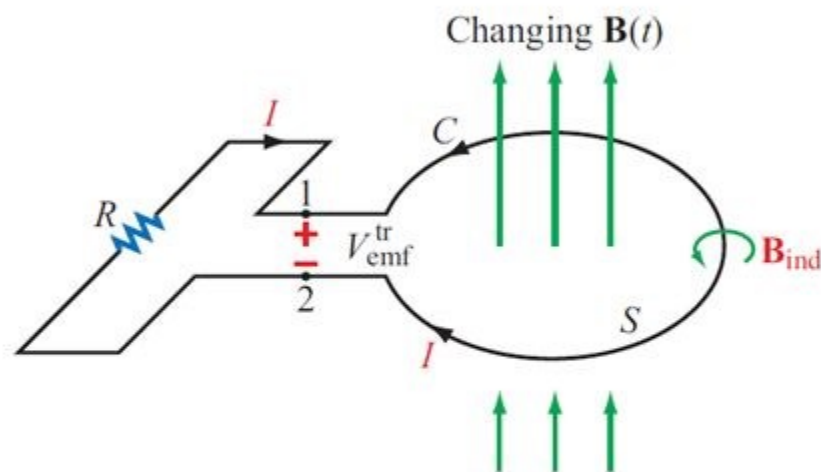
$$V_{\text{emf}}^{\text{tr}} = -N \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s} \quad (\text{transformer emf}),$$

The connection between the direction of $d\mathbf{s}$ and the polarity of $V_{\text{emf}}^{\text{tr}}$ is governed by the following right-hand rule: if $d\mathbf{s}$ points along the thumb of the right hand, then the direction of the contour C indicated by the four fingers is such that it always passes across the opening from the positive terminal of $V_{\text{emf}}^{\text{tr}}$ to the negative terminal.

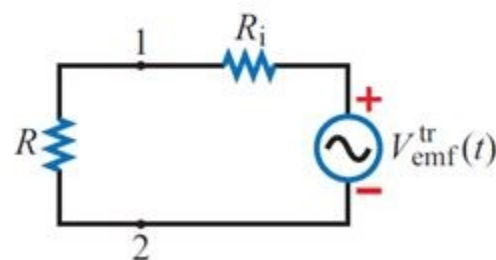
$$I = \frac{V_{\text{emf}}^{\text{tr}}}{R + R_i}. \quad (6.9)$$

For good conductors, R_i usually is very small, and it may be ignored in comparison with practical values of R .

*The polarity of $V_{\text{emf}}^{\text{tr}}$ and hence the direction of I is governed by **Lenz's law**, which states that the current in the loop is always in a direction that opposes the change of magnetic flux $\Phi(t)$ that produced I .*



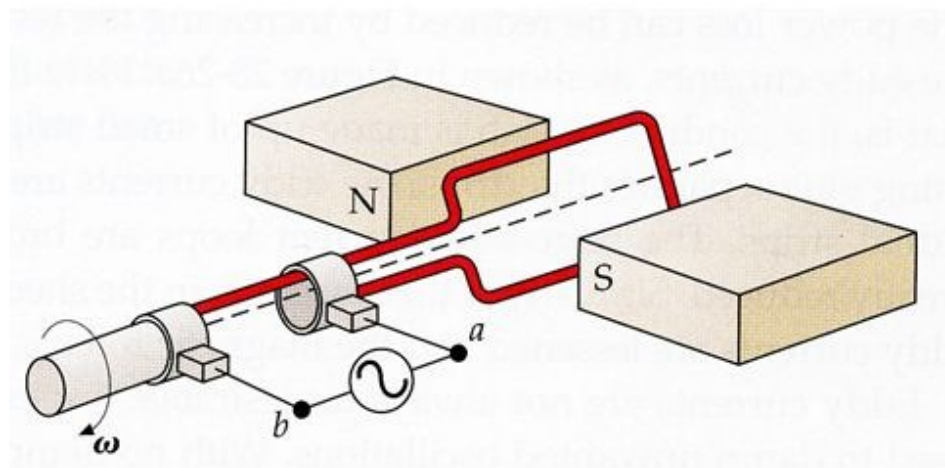
(a) Loop in a changing $\mathbf{B}$ field



(b) Equivalent circuit

Figure 6-2: (a) Stationary circular loop in a changing magnetic field $\mathbf{B}(t)$, and (b) its equivalent circuit.

Electric Generators (AC)

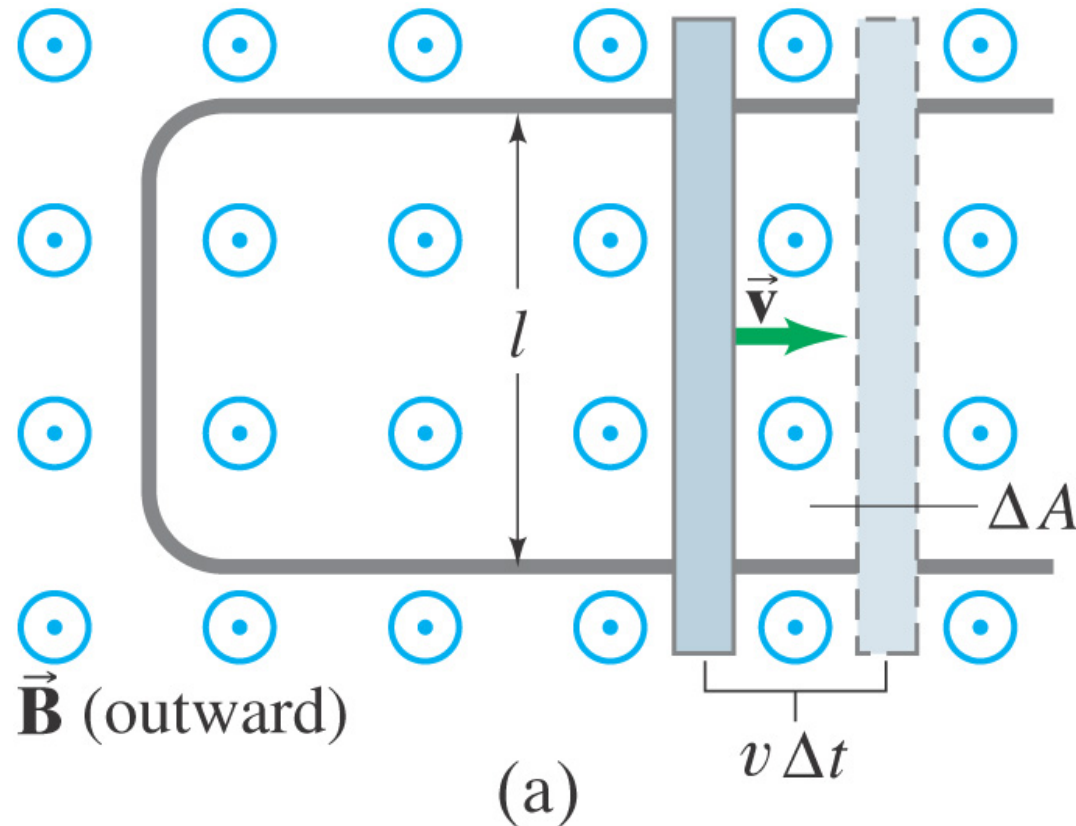


- A mechanically turned armature, consisting of N-loops of a coil wound on the armature
- As the armature is rotated, an EMF is induced in the rotating coil, which alternates as the magnetic flux through the loop reverses. The induced EMF is:

$$E = NBA \sin(\omega t) \text{ or } E_0 \sin(\omega t) \text{ where } E_0 = NBA$$

21.3 EMF Induced in a Moving Conductor

This image shows another way the magnetic flux can change:

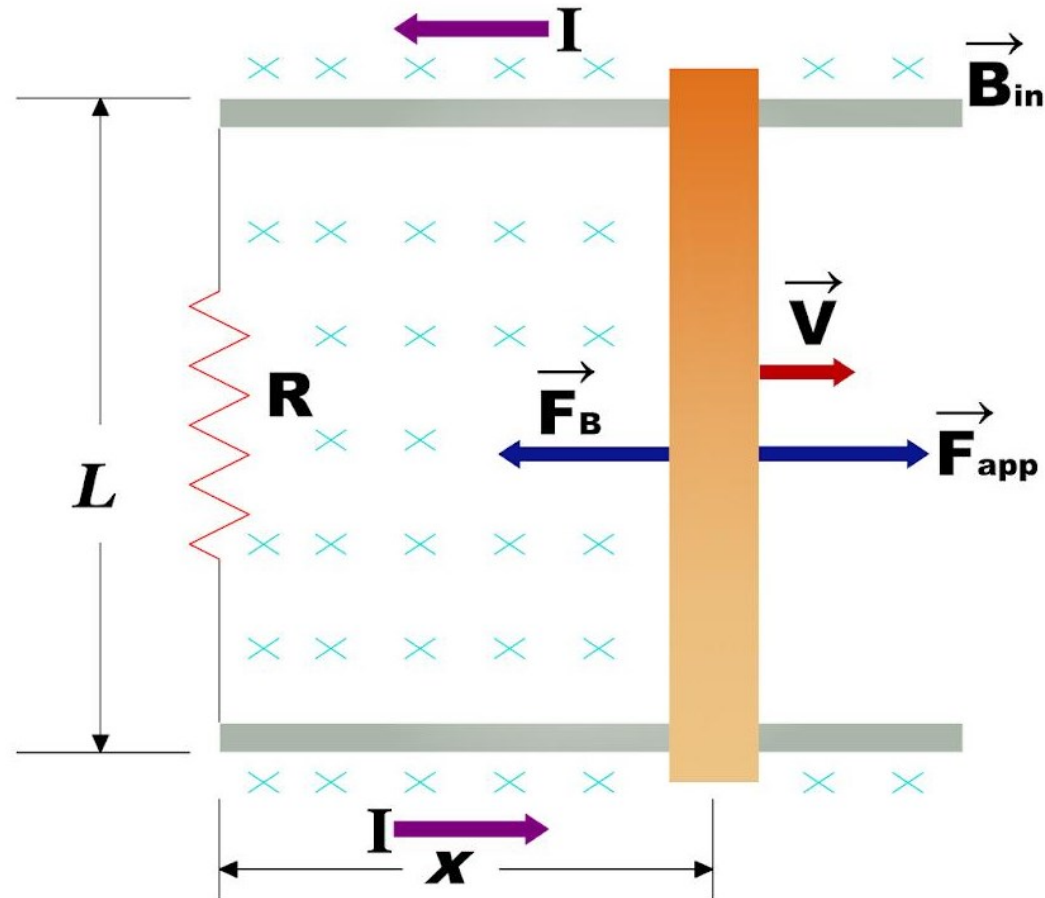


Sliding Rod in a Magnetic Field

A conducting rod of length L slides to the right with constant velocity v on two parallel rails. The rails are connected by a resistor R . A uniform magnetic field B is directed into the page.

👉 Find:

1. Induced EMF
2. Induced current
3. Direction of current



Step 1: Use Faraday's Law

$$\mathcal{E} = -\frac{d\Phi_B}{dt}$$

Magnetic flux:

$$\Phi_B = B \cdot A$$

Area of loop:

$$A = L \cdot x(t)$$

So:

$$\Phi_B = BLx$$

Step 2: Differentiate Flux

$$\frac{d\Phi_B}{dt} = BL \frac{dx}{dt}$$

But:

$$\frac{dx}{dt} = v$$

So:

$$\frac{d\Phi_B}{dt} = BLv$$

⚡ Step 3: EMF

$$\mathcal{E} = -BLv$$

Magnitude:

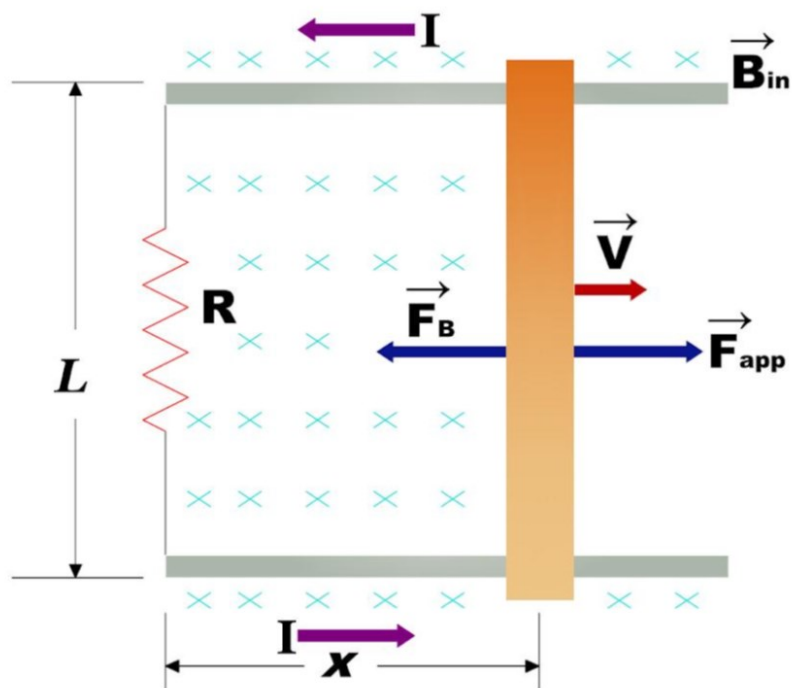
$$|\mathcal{E}| = BLv$$

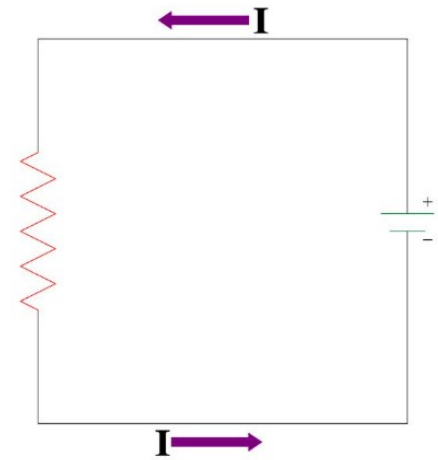
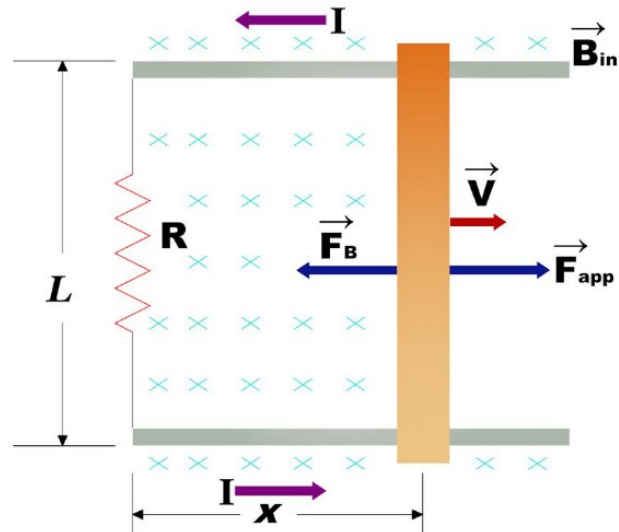
🔑 Step 4: Current

Using Ohm's law:

$$I = \frac{\mathcal{E}}{R}$$

$$I = \frac{BLv}{R}$$





Step 5: Direction (Lenz's Law)

- Magnetic field is **into the page**
- Area is increasing $\rightarrow$ flux **increases**
- Induced current must **oppose** increase

👉 So induced field must be **out of the page**

Using right-hand rule:

Current is counterclockwise

Final Answers

$$\mathcal{E} = BLv$$

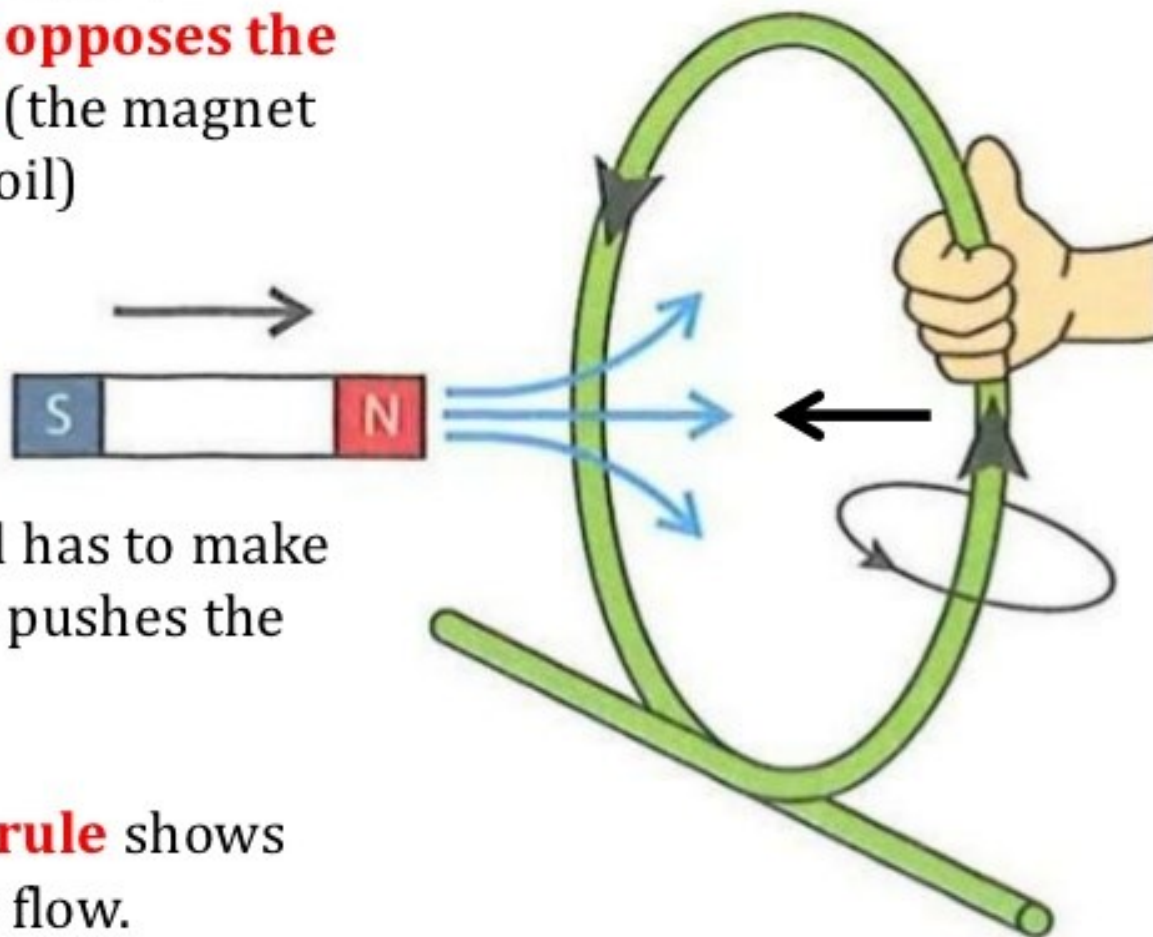
$$I = \frac{BLv}{R}$$

Direction: counterclockwise

Faraday's and Lenz's Laws – Example 1

B) Lenz's Law

The direction of the Induced current is such that **it opposes the change producing it** (the magnet moving towards the coil)

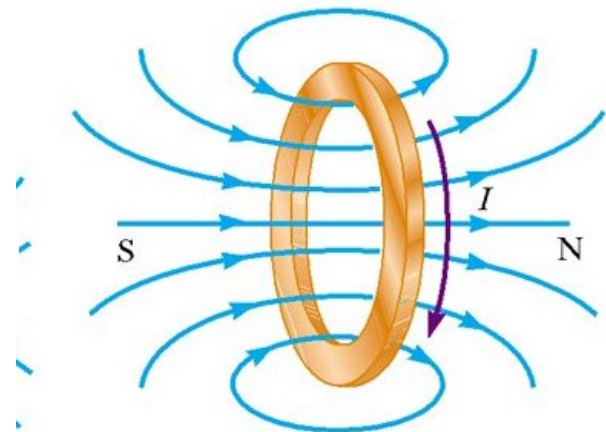
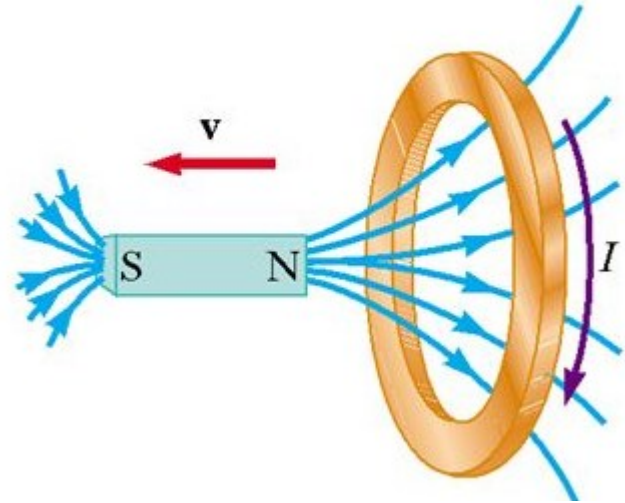
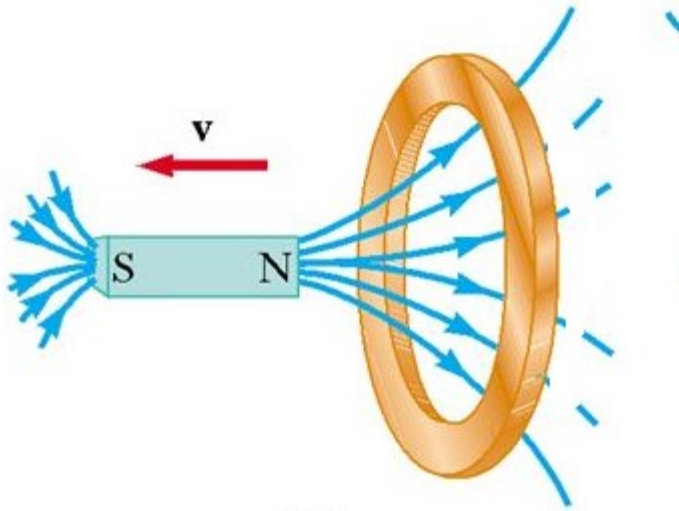


The current in the coil has to make a **magnetic field** that pushes the Magnet away....

The **Right hand grip rule** shows how the current must flow.

Retrieval Session

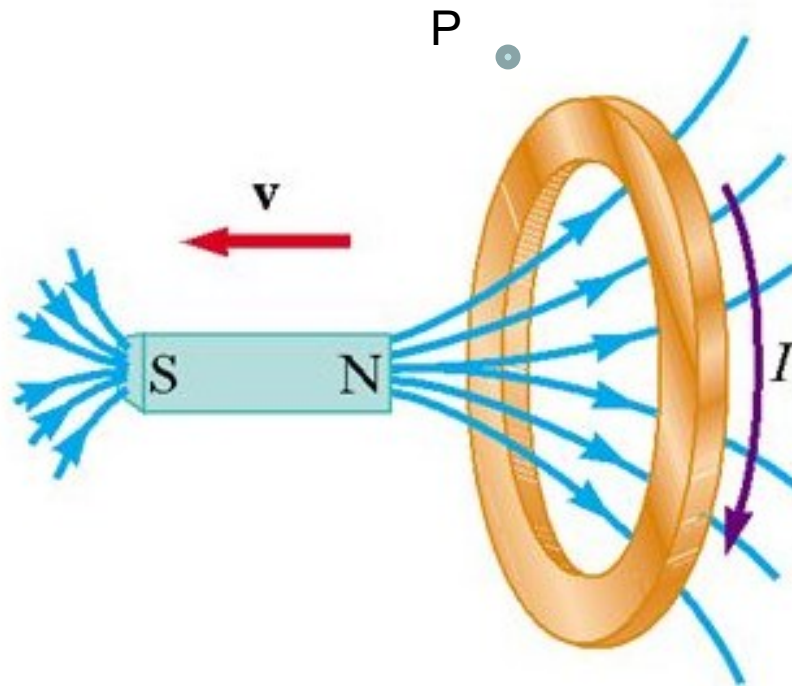
What is the direction of the induced current?



Please type: C (Clockwise) or CC (counterclockwise)

Retrieval Session

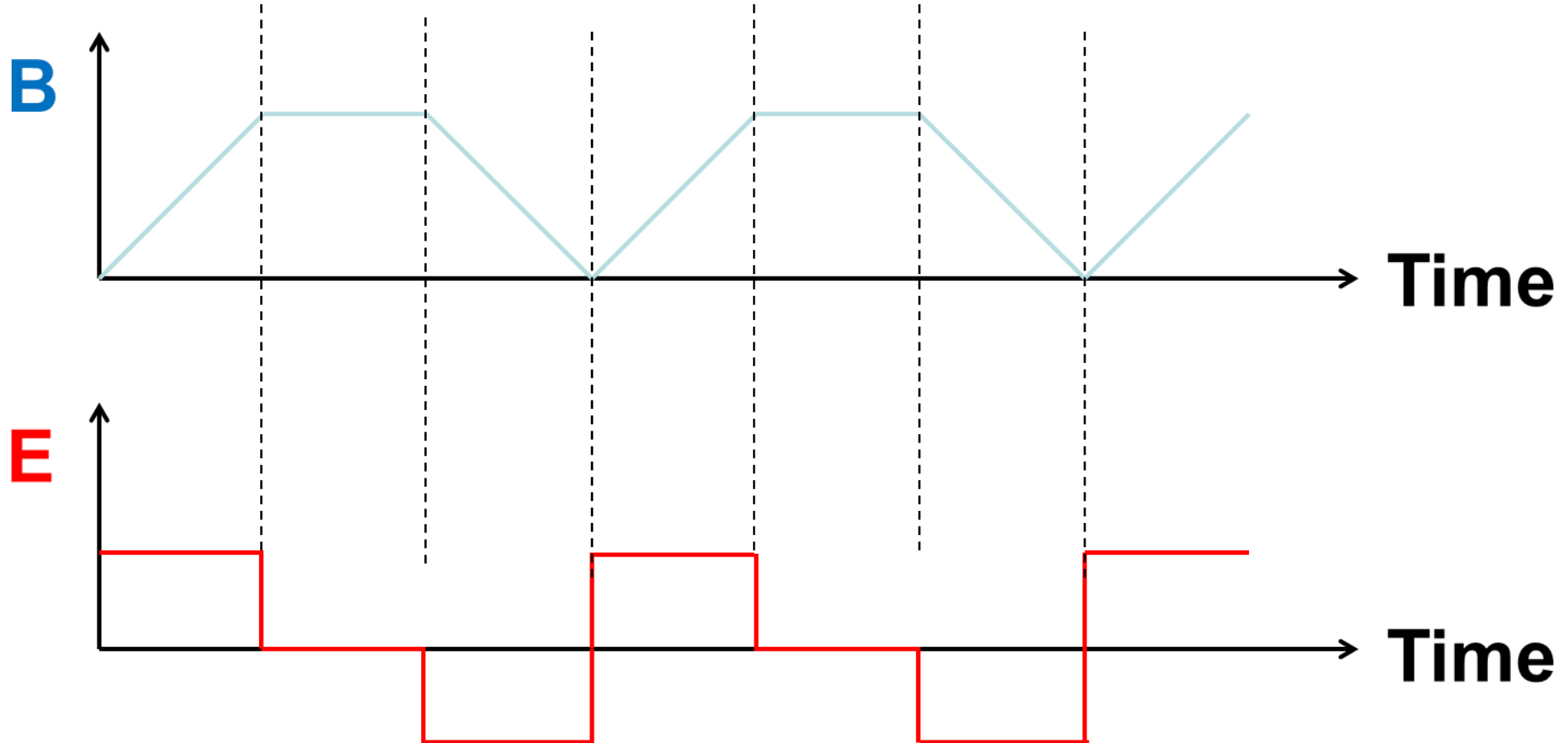
What is the electric field in point P ?



Please type: zero or Not zero

Retrieval Session

- Does time-varying magnetic field always produce time-varying electric field?



Static vs Time-Varying Electric Field

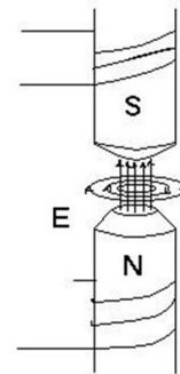
Static electric field

- ❑ Produced by static electric charges
- ❑ Electric field lines start on the positive charge(s) and end on negative charge(s)
- ❑ The work done moving charge on a closed contour is zero

$$\oint_L \vec{E} \cdot d\vec{l} = \oint_L E dl \cos \alpha = \oint_L E_l dl \equiv 0$$

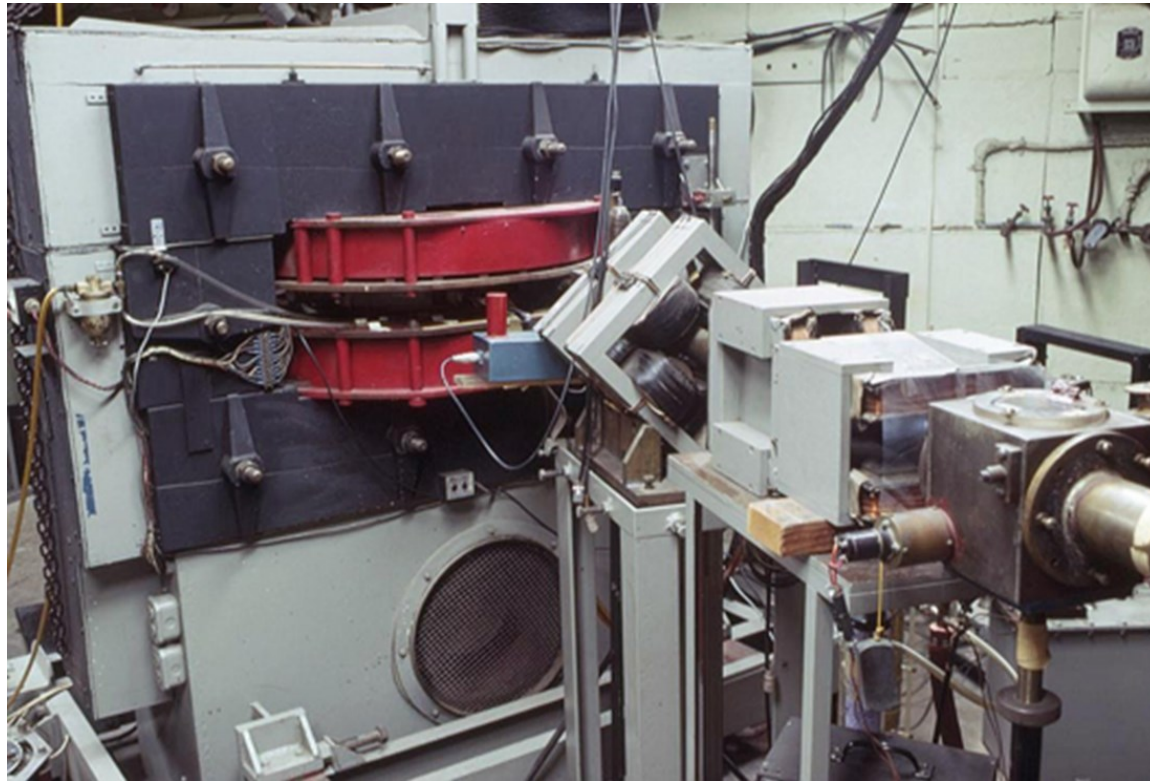
Time-varying electric field

- ❑ Produced by time-varying magnetic field
- ❑ Electric field lines are closed contours
- ❑ The work done moving charge on a closed contour may be NOT zero



$$\oint_c \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d}{dt} \int_s \mathbf{B} \cdot d\mathbf{s}$$

The Betatron



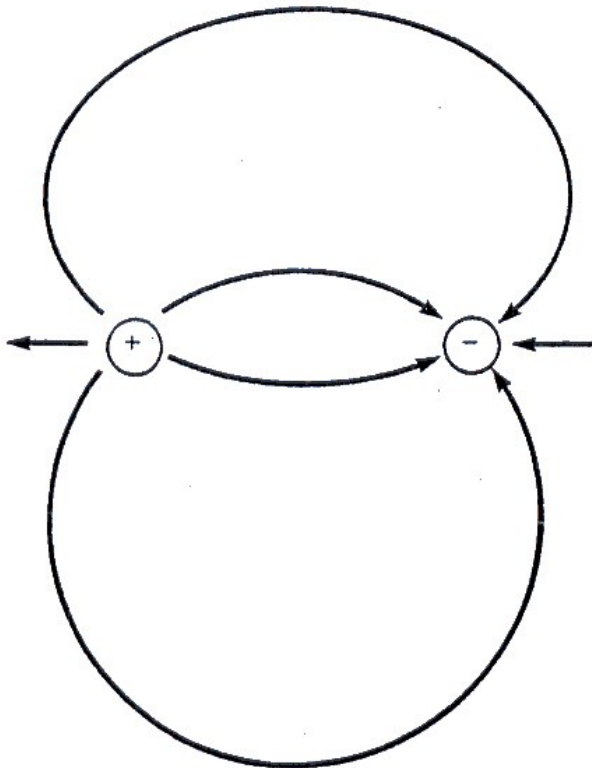
One of the first betatrons, built in the early 1940s. The so-called 20 inch machine at the University of Illinois.

<https://www.youtube.com/watch?v=LEHrUcoQjx8>

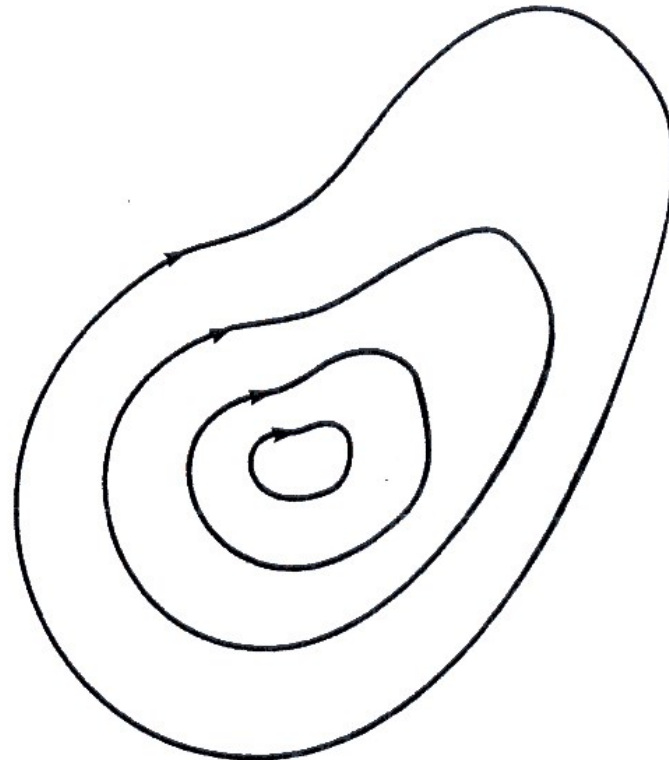
Different forms of Electric Field

Which of the graphs (left or right) shows electric field?

Electric field produced by charges



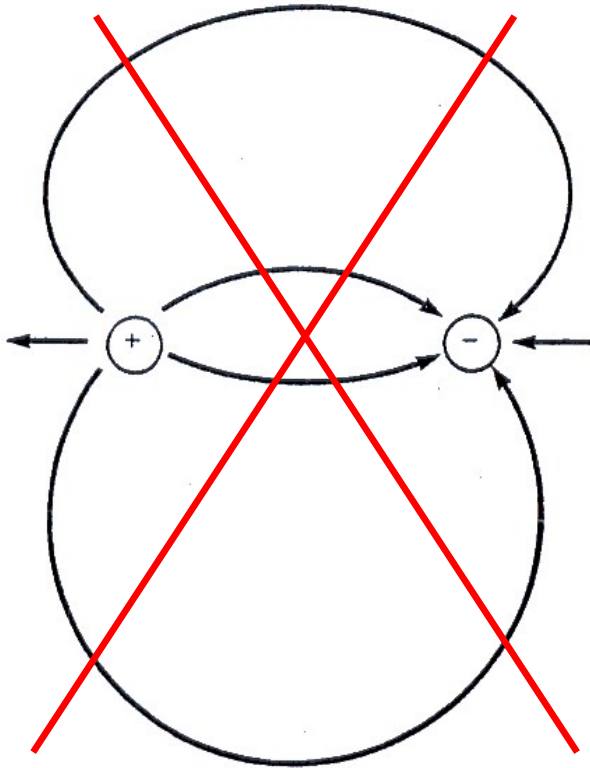
Vortex electric field (e.g. produced by time varying magnetic field)



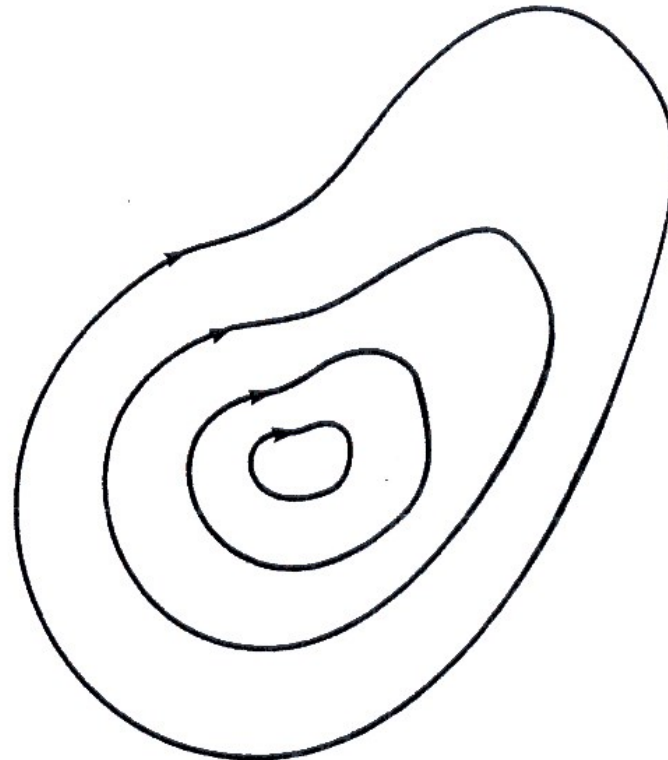
Different forms of Magnetic Field

Which of the graphs (left or right) shows magnetic field?

There are no magnetic charges!



Vortex magnetic field
(e.g. produced by time
varying electric field)

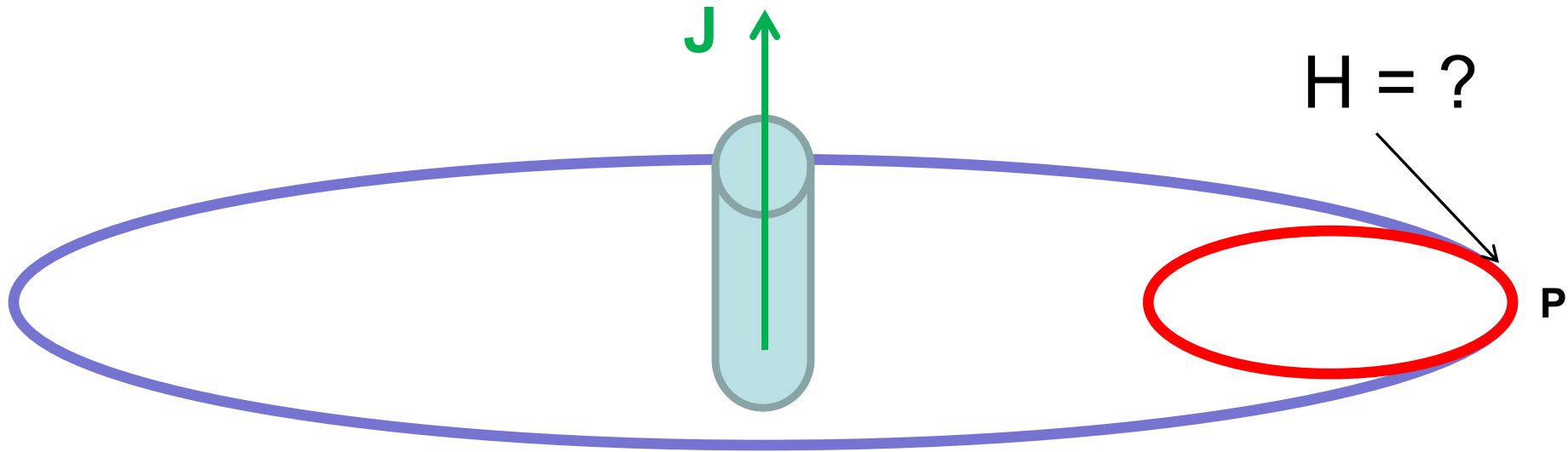


Ampere's Law

4. Ampere's Law

$$\oint_C \frac{\vec{B}}{\mu_0} \cdot d\vec{\ell} = \int_S \vec{J} \cdot d\vec{S} + \frac{d}{dt} \int_S \epsilon_0 \vec{E} \cdot d\vec{S}$$

Retrieval Session

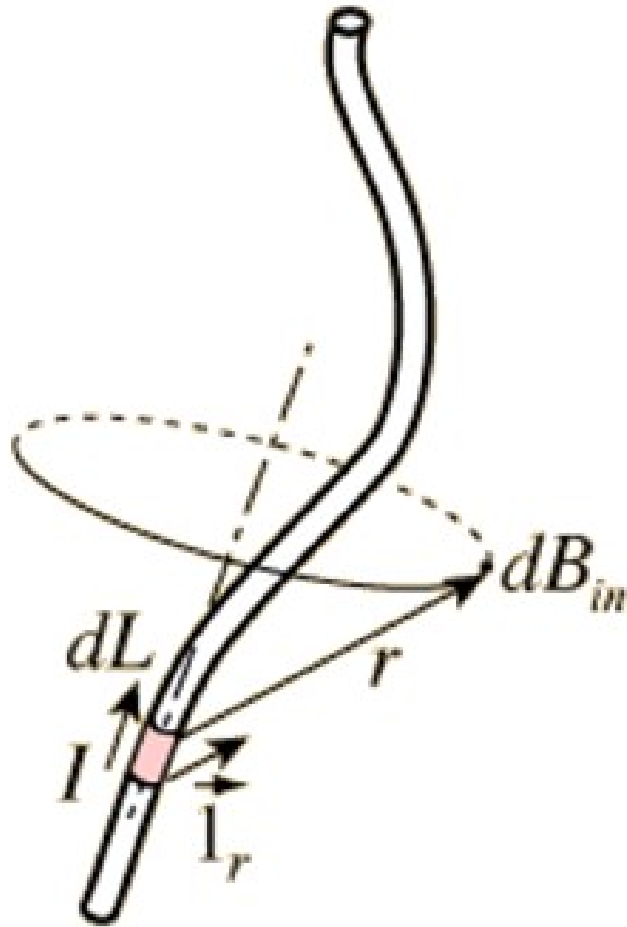


Does magnetic field
depend on the contour
we choose?

$$\oint \vec{H} \cdot d\vec{l} = \int \vec{J} \cdot d\vec{s}$$

$$\oint \vec{H} \cdot d\vec{l} = 0$$

The Biot-Savart law



Magnetic field
of a current
element

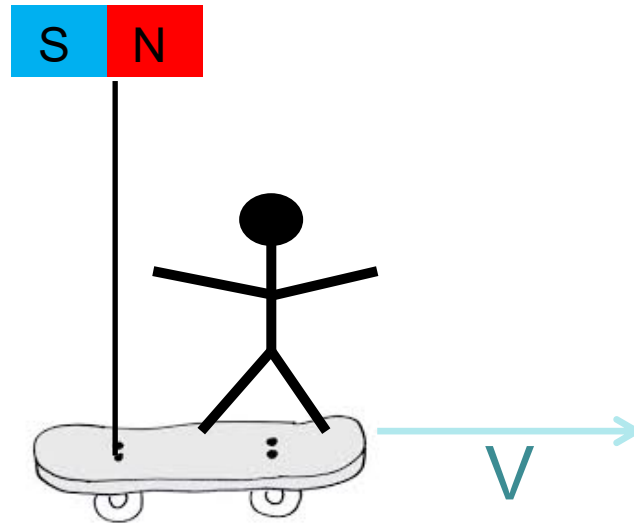
$$d\vec{B} = \frac{\mu_0 I d\vec{L} \times \hat{l}_r}{4\pi r^2}$$

where

$d\vec{L}$ = infinitesimal length of conductor
carrying electric current I

$\hat{l}_r$ = unit vector to specify the direction
of the the vector distance r from
the current to the field point.

EM Fields and Relativity Theory



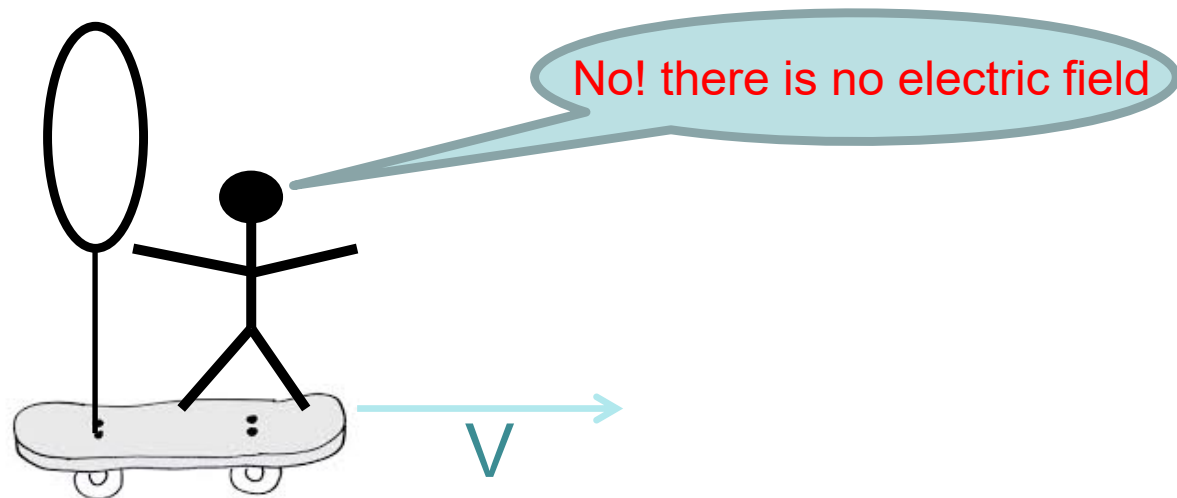
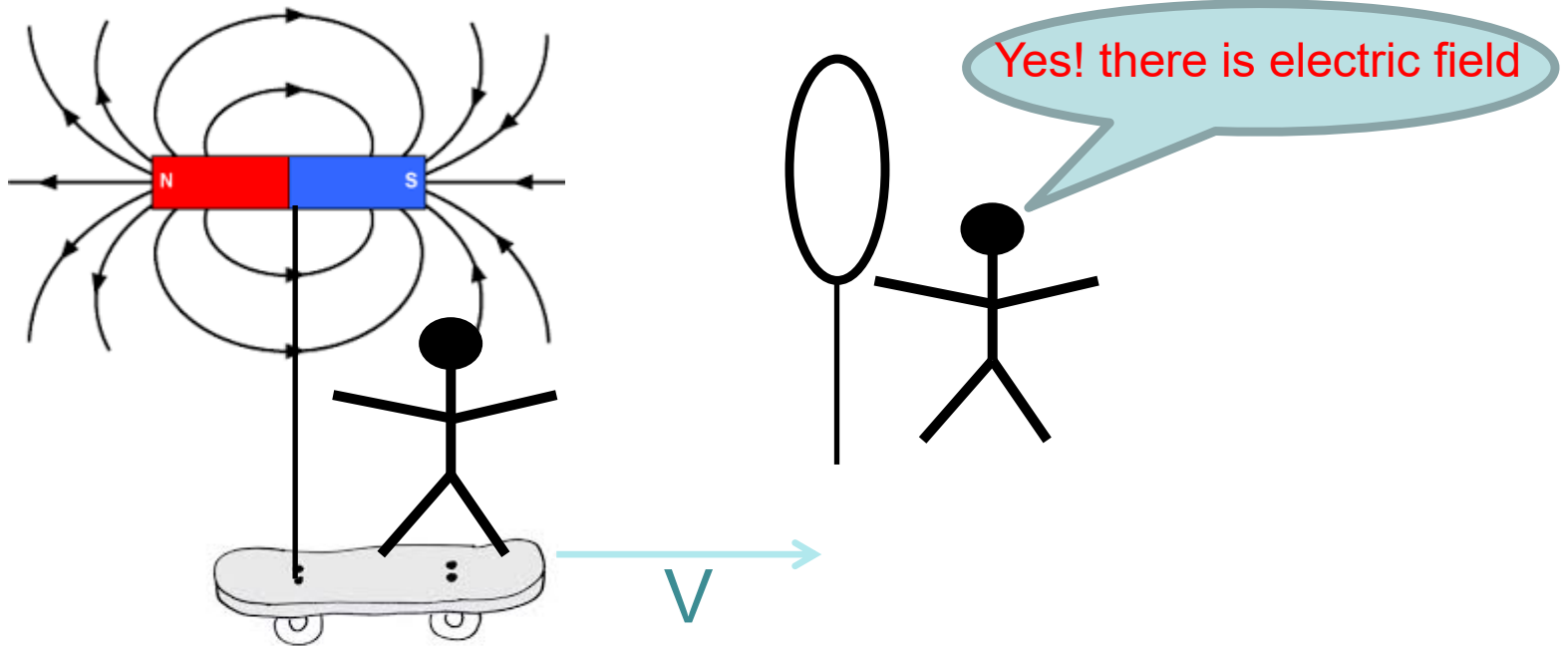
Hmm.. Does EM fields depend on the frame of reference?

Let us consider a person on a skateboard moving with constant speed V

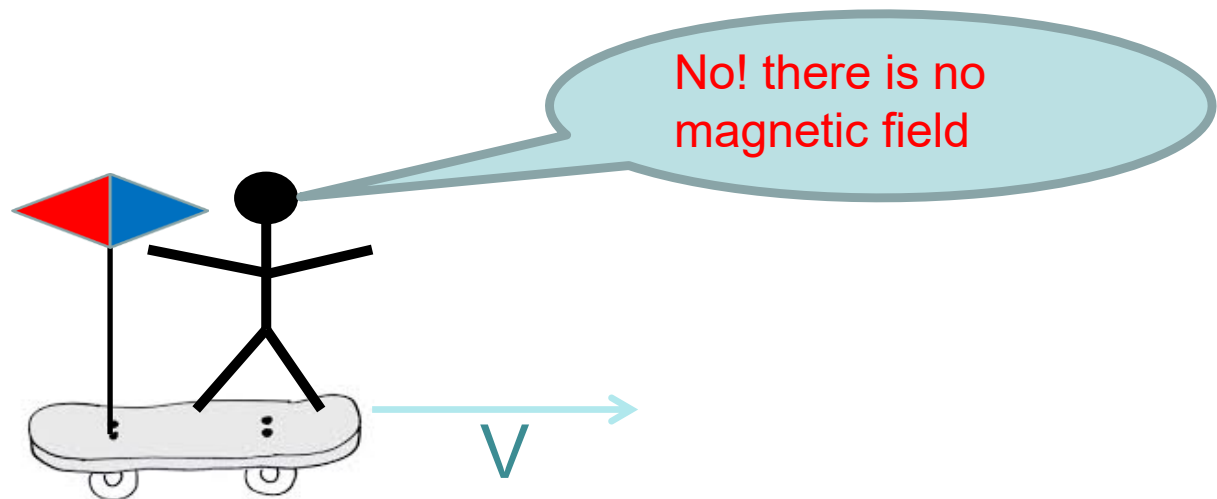
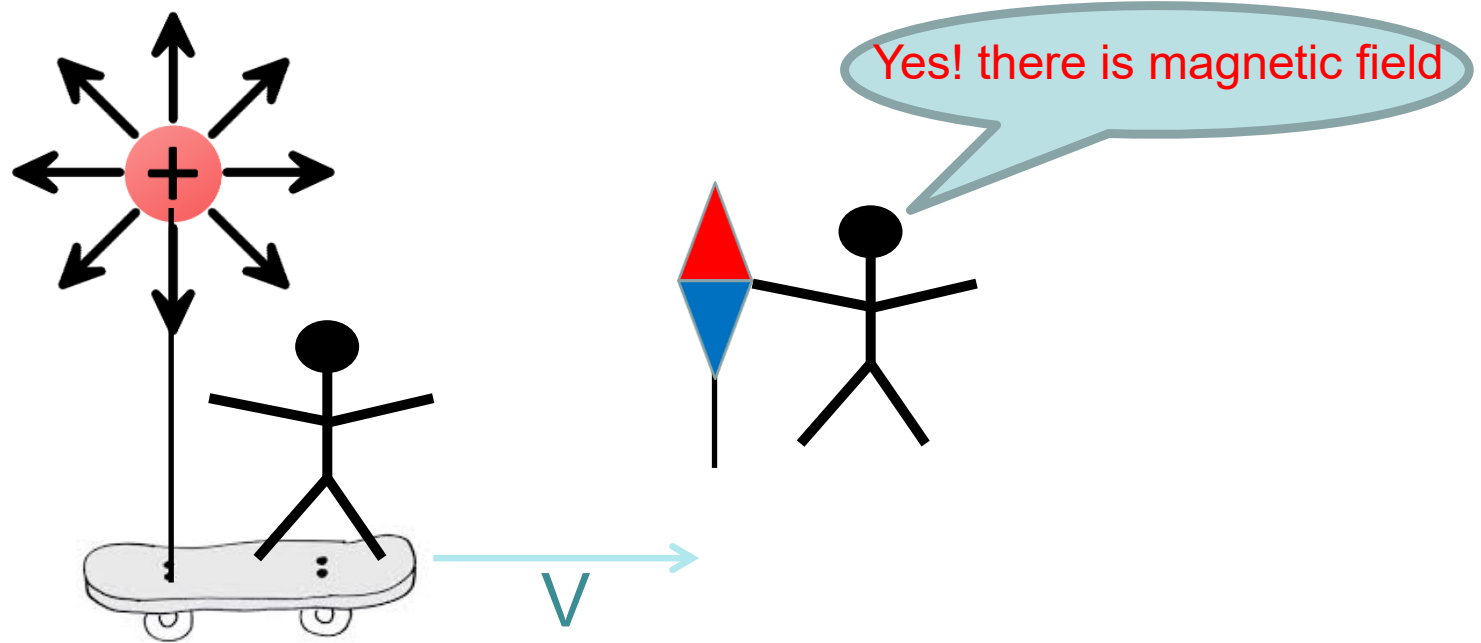
There is a permanent magnet attached to the skateboard

Is there an electric field ?

Electric field depends on the frame of reference



Magnetic field depends on the frame of reference



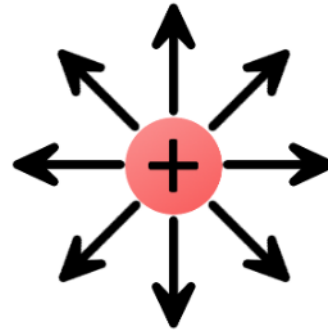
What is magnetic field now?

Compass as a watch



It is + 20.5 G

V



It is + 20.5 G



V



It is 0 G,
I missed my train

Postulates of the Special Theory of Relativity

- **First Postulate:** *The laws of physics have the same form in all inertial reference systems. (This is the Relativity Principle)*
- **Second Postulate:** *Light propagates through empty space with a definite speed “ c ” independent of the speed of the source or of the observer. (Agrees with experiment)*

Relativistic Electromagnetism

- The requirement that the equations remain consistent when viewed from various moving observers led to special relativity, a geometric theory of 4-space where intermediation is by light and radiation. The spacetime geometry provided a context for technical description of electric technology, especially generators, motors, and lighting at first. The Coulomb force was generalized to the Lorentz force. For example, with this model transmission lines and power grids were developed and radio frequency communication explored.

$$E'_x = E_x$$

$$B'_x = B_x$$

$$E'_y = \frac{(E + v \times B)_y}{\sqrt{1 - v^2}}$$

$$B'_y = \frac{(B - v \times E)_y}{\sqrt{1 - v^2}}$$

$$E'_z = \frac{(E + v \times B)_z}{\sqrt{1 - v^2}}$$

$$B'_z = \frac{(B - v \times E)_z}{\sqrt{1 - v^2}}$$